Vincent Griffo

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EDUCATION

Georgia Institute of Technology, Atlanta, GA — *Expected: May 2027*
Bachelor of Science in Mechanical Engineering; Minor in Physics; Concentration in Robotics — *GPA: 4.00/4.00*

Harrison High School, Harrison, NY — *June 2022*
High School Diploma: Summa Cum Laude — *GPA: 4.00/4.00*

AWARDS

Tohoku Short-Term Trainee — *May 2026*
Selected from an applicant pool by Tohoku University's School of Engineering to conduct research in Japan under a faculty member of my choice.

Astronaut Scholarship — *May 2026*
National scholarship awarded to top undergraduate researchers demonstrating exceptional academic achievement

Georgia Tech Capstone Honorable Mention — *April 2026*
Awarded Honorable Mention for *Sidewalk Sidekick* during the Georgia Tech Spring 2026 Capstone Design Expo, recognizing one of the top 8 non-winning teams among more than 200 capstone teams university-wide.

Goldwater Scholar — *March 2026*
Competitive merit scholarship awarded to undergraduate students with strong research experience and academics

RSS Space Robotics Workshop Best Paper — *June 2025*
Received Best Paper for GROVER (variable-height grouser wheel) at the RSS Space Robotics Workshop 2025

ThinkSwiss Research Fellowship — *February 2025*
Recipient of a competitive award to research at ETH Zurich under Dr. Marco Hutter of the Robotic Systems Lab

President's Undergraduate Salary Award x2 — *November 2024, November 2025*
Awarded a competitive research grant to support research at AstroX

University Rover Challenge Final Round x2 — *March 2024, March 2025*
Rover advanced to the final round of the University Rover Challenge, competing against 32 international teams at the Mars Desert Research Station in Hanksville, Utah

Stamps President's Scholar — *August 2022*
Full-ride merit scholarship awarded to the top 1% of incoming students at Georgia Tech based on commitment to scholarship, leadership, progress, and service

RESEARCH

AstroX | Georgia Tech — *August 2024 – Present*
Principal Investigator: Prof. Yashwanth K. Nakka

Design, Control, and Testing of a Highly Dexterous Multimodal Rover

- Lead a team of seven engineers designing an 8-DOF multimodal rover leg to experimentally study how leg dexterity and morphology influence locomotion efficiency and stability on lunar analog terrain
- Designed low-inertia serial leg architectures using inverse dynamics with MuJoCo and Pinocchio to evaluate joint torque requirements and actuator feasibility across multiple gait morphologies
- Integrated mechanical differentials and transmission strategies to reduce reflected inertia and peak joint torques during multimodal gait transitions

Design, Control, and User Testing of a Custom Wheeled Quadruped for Blind and Low-Vision Individuals

- Led a multidisciplinary team of six engineers while serving as the project's sole mechanical and controls engineer, developing a 12-DOF (3-DOF/leg) wheeled quadruped capable of 2.7 m/s rolling and 1.05 m/s walking
- Designed the complete mechanical system in SolidWorks, performing inverse dynamics in MuJoCo for actuator sizing and engineering high-load shafts, bearings, transmissions, and structural members validated through finite element analysis on seven machined aluminum components
- Developed the robot's control architecture using rigid-body dynamics, impedance control, dynamic gait generation, CAN bus communication, and EKF-based state estimation for robust multimodal locomotion

- Integrated a Jetson Nano autonomy stack implementing YOLO-based computer vision, obstacle detection, and A* path planning for autonomous navigation and obstacle avoidance
- Manufactured and assembled custom components through CNC machining and waterjet fabrication, culminating in full-system integration and user testing with blind and low-vision participants

Design of Rover Mobility Fleet for Multi-Agent Collaboration (NOVA)
- Led the design of a low-cost collaborative rover fleet for multi-agent autonomy experiments across AstroX's lunar terrain testbed to inform future lunar cobot missions
- Designed and manufactured the full rover mechatronic system, including chassis, rocker suspension, and four independent swerve-drive modules enabling omnidirectional motion at speeds up to 2 m/s
- Architected the complete 12 V electrical system, including battery specification, power distribution, actuator selection, relays, and dual Jetson Nano integration
- Integrated dual ZED Mini stereo cameras and onboard perception to enable cooperative navigation and collaborative environment understanding across multiple rovers

Design and Control of Variable-Height Grouser Platform for Extraterrestrial Navigation
- Invented and designed an internal spiral cam mechanism within a wheel that dynamically adjusts grouser height to match analog Martian terrain properties, achieving up to a 58% reduction in slip ratio in granular media and informing how adaptability may aid crucial locomotion in future Martian missions
- Performed structural FEA in SOLIDWORKS on spiral cam and tread geometries, informing stiffness, contact stress, and fatigue limits under dynamic terrain interaction loads
- Designed a fully instrumented terrain testing gantry with linear and rotary encoders, current and voltage sensing to experimentally quantify slip ratio, tractive force, and energy consumption across four terrains
- Developed a closed-loop PID controller in Python to precisely regulate grouser geometry, enabling accurate and repeatable experimental validation of wheel–terrain interaction

Space Infrastructure Engineering Lab | Tohoku University, Japan *May 2026 – July 2026*
Principal Investigator: Prof. Toshinori Kuwahara

Design and Control of Electromechanical Soft Docking System for Floating Platform Cubesat
- Designed and integrated a mountable soft-capture docking mechanism with complementary male and female interfaces for autonomous CubeSat rendezvous, capture, and mechanical locking.
- Designed a servo-actuated twist-lock mechanism with ball-detent alignment features, enabling low-friction docking and repeatable locking and undocking operations.
- Designed, wound, and optimized a custom iron-core electromagnet from first principles to enable long-range electromagnetic soft capture on the Space Infrastructure Engineering Lab's air-bearing test platform.
- Applied electromagnetic first principles and a current-controlled test system to maximize magnetic flux while minimizing thermal losses, achieving reliable capture distances of up to 37 cm.

Robotic Systems Lab / LunarLeaper | ETH Zurich, Switzerland *August 2025 – December 2025*
Principal Investigator: Prof. Marco Hutter

Design and Control of a 3-DOF Parallel, Space-Grade Arachnid Robot (LunarLeaper)
- Led mission-driven mechanical design of a 3-DOF parallel leg mechanism for *LunarLeaper*, a 12U CubeSat-class quadrupedal robot planned for a 2030 lunar science exploration mission
- Iterated through six full-scale leg prototypes using rapid prototyping and COTS hardware to converge on a compact architecture capable of surviving lunar dust, thermal cycling, and launch constraints
- Executed full mechatronic development of the final prototype using PMSM actuators, employing stationary actuator placement to eliminate moving thermal straps and wiring, improving system lifetime
- Leveraged joint torque and ROM analysis in Simscape Multibody to implement final transmission architectures using heim-joint yaw and hip pitch joints and a steel-shaft flexure knee mechanism

Aerial Robotics and Experimental Autonomy Lab | Georgia Tech *August 2023 – August 2024*
Principal Investigator: Prof. Jonathan Rogers

Design and Control of Multimodal UAS Proprotor Articulation for VTOL-to-Fixed-Wing Transition

- Worked on a novel proprotor articulation mechanism to allow seamless transition between vertical take-off and landing (VTOL) and fixed-wing flight modes for a scaled-down unmanned aerial vehicle (UAV)
- Created mathematical models and performed simulations of articulation using MATLAB, applying principles of rigid body mechanics to optimize spatial constraints, maximize torque output, and enhance moment arm rotation
- Designed and implemented a PID control system in MATLAB Simulink, integrating hardware-in-the-loop (HIL) simulation with a DC motor and Arduino Nano for application in the UAS control scheme

Laboratory for Intelligent Decision and Autonomous Robots | Georgia Tech *December 2022 – March 2023*
Principal Investigator: Prof. Ye Zhao

Development of Control Systems and Autonomous Navigation for Human-Avoidance Drones

- Leveraged ROS and Python to design autonomous human-avoidance algorithms to enable real-time drone navigation and obstacle detection for enhanced flight path control

SSP Astrophysics | Online through MIT, Caltech, and Harvey Mudd Sponsorship *May 2021 – July 2021*
Principal Investigators: Prof. Michael Hannawald & Prof. Cassandra Fallscheer

Tracking, Modeling, and Orbital Prediction of Near-Earth Asteroid 2003 HB6

- Determined the position of an asteroid relative to known stars using parallax and coordinate transformations with data from industry-grade telescopes at observatories in Chile, Australia, and the USA
- Calculated orbital parameters of asteroids utilizing the Method of Gauss and Monte Carlo simulation to be used for a precise visual model and accurate approximations within 0.1% of JPL's previously predicted orbital elements
- Developed a mathematical model and 3D visual simulation using Python to predict asteroid positions decades in advance based on determined orbital parameters
- Collaborated with the Southwest Research Institute to analyze asteroid collision history using advanced orbital simulations developed with SWIFT

JOURNAL PUBLICATIONS

[1] **V. Griffo**, Y. Nakka, Terrain-Adaptive Grouser Wheel for Optimal Planetary Exploration: Design and Experimental Investigation **(Submitted IEEE T-FR, revisions requested)**.

[2] **V. Griffo**, B. Ma, J. Yu, Data on Orbit Determination of Asteroid 326732 (2003 HB6)
(Internal to the Harvard Minor Planet Center).

CONFERENCE PUBLICATIONS

[1] Y. Nakka, **V. Griffo**, S. Kumar, C. DelGandio, T. Dokania, A. Bhardwaj, S. Kelkar, U.K. Ponugupati. A Lunar Terrain Facility for Robust and Resilient Learning-Based Autonomy. *IEEE Aerospace Conference*, Big Sky, MT, USA, March 2027. **(Submitted)**

[2] J. Song, **V. Griffo**, A. Gandhi, Y. Nakka. Hierarchical CS-MPC for Terrain-aware Mode-switching and Control. *Lunar Surface Innovation Consortium*, Atlanta, GA, USA, October 2026. **(Accepted abstract)**

[3] **V. Griffo**, Y. Nakka. Terrain-Adaptive Grouser Wheel for Optimal Planetary Exploration: Design and Experimental Investigation. *IEEE IROS ROSE*, Pittsburgh, PA, USA, October 2026.

[4] S. Stähler, A. Mittelholz, A. Cocheril, A. Fuhrer, J. Church, E. Krasnova, H. Kolvenbach, J. Smajic, V. Bickel, M. Betschen, E. Brändli, J. Gomez Jodar, **V. Griffo**, L. Heyun, N. Hirzel, Y. Hofmann, J. Isler, Y. Luginbühl, J. Majer, R. Margarit, G. Smith-Vaniz, M. Hutter, R. Kornfeld, LunarLeaper: A university-led legged robotic mission to the moon. *5th Symposium on Space Educational Activities*, Ottobrunn–Taufkirchen, Germany, April 2026.

[5] **V. Griffo**, Y. Nakka, Revised Development and Adaptive Control of a Multimodal Wheel for Planetary Exploration. *Lunar Surface Innovation Consortium*, Atlanta, GA, USA, November 2025. **(Accepted, Canceled due to Government Shutdown)**

[6] S. Kumar, **V. Griffo**, Y. Nakka, NOVA: Design Overview of an Autonomous 3D-printed Rover Platform. *Lunar Surface Innovation Consortium*, Atlanta, GA, USA, November 2025. **(Accepted, Canceled due to Government Shutdown)**

[7] **V. Griffo**, Y. Nakka, Design and Control of a Multimodal Wheel for Adaptable Extraterrestrial Mobility. *Robotics: Science and Systems Space Robotics*, Los Angeles, CA, USA, June 2025. **(Best Paper)**

ORAL PRESENTATIONS

[1] **V. Griffo**, Adaptive Robotics and Multimodal Exploration, *Astronaut Scholarship Innovators Symposium*, Houston, TX, USA, August 2026.

[2] **V. Griffo**, T. Kuwahara, Design and Control of an Electromechanical Soft-Docking System for Floating-Platform CubeSats. *Tohoku University LQS*, Sendai, Japan, July 2026.

[3] **V. Griffo**, S. Kumar, A. Espy, S. Sahas, P. Nair, K. Padmanabhan, Design, Control, and User Testing of a Custom Wheeled Quadruped for Blind and Low-Vision Individuals. *Georgia Institute of Technology Capstone Design Expo*, Atlanta, GA, USA, April 2026.

[4] **V. Griffo**, Y. Nakka, Revised Development and Adaptive Control of a Multimodal Wheel for Planetary Exploration. *Lunar Surface Innovation Consortium*, Atlanta, GA, USA, November 2025. **(Accepted, Canceled due to Government Shutdown)**

[5] **V. Griffo**, Y. Nakka, Design and Control of a Multimodal Wheel for Adaptable Extraterrestrial Mobility. *Robotics: Science and Systems Space Robotics*, Los Angeles, CA, USA, June 2025.

[6] **V. Griffo**, S. Kumar, Y. Nakka, GROVER: A Variable Height Grouser Wheel for Autonomous Enhanced Terrain Adaptability. *Georgia Tech Annual Research Symposium*, Atlanta, GA, USA, April 2025.

POSTER PRESENTATIONS

[1] J. Song, **V. Griffo**, A. Gandhi, Y. Nakka. Hierarchical CS-MPC for Terrain-aware Mode-switching and Control. *Lunar Surface Innovation Consortium*, Atlanta, GA, USA, October 2026. **(Accepted abstract)**

[2] **V. Griffo**, Y. Nakka. Terrain-Adaptive Grouser Wheel for Optimal Planetary Exploration: Design and Experimental Investigation. *IEEE IROS ROSE*, Pittsburgh, PA, USA, October 2026.

[3] **V. Griffo**, Y. Nakka. Terrain-Adaptive Grouser Wheel for Optimal Planetary Exploration: Design and Experimental Investigation. *GT Space Week*, Atlanta, GA, USA, September 2026.

[4] **V. Griffo**, S. Kumar, A. Espy, S. Sahas, P. Nair, K. Padmanabhan, Design, Control, and User Testing of a Custom Wheeled Quadruped for Blind and Low-Vision Individuals. *Georgia Institute of Technology Capstone Design Expo*, Atlanta, GA, USA, April 2026.

[5] **V. Griffo**, Y. Nakka, Design and Control of a Multimodal Wheel for Adaptable Extraterrestrial Mobility. *Lunar Surface Innovation Consortium*, Atlanta, GA, USA, November 2025. **(Accepted, Canceled due to Government Shutdown)**

[6] S. Kumar, **V. Griffo**, Y. Nakka, NOVA: Design Overview of an Autonomous 3D-printed Rover Platform. *Lunar Surface Innovation Consortium*, Atlanta, GA, USA, November 2025. **(Accepted, Canceled due to Government Shutdown)**

[7] **V. Griffo**, Y. Nakka, Design and Control of a Multimodal Wheel for Adaptable Extraterrestrial Mobility. *Robotics: Science and Systems Space Robotics*, Los Angeles, CA, USA, June 2025.

[8] **V. Griffo**, S. Kumar, Y. Nakka, GROVER: A Variable Height Grouser Wheel for Autonomous Enhanced Terrain Adaptability. *Georgia Tech College of Sciences EXPLORE*, Atlanta, GA, USA, March 2025.

[9] **V. Griffo**, S. Kumar, Y. Nakka, GROVER: A Variable Height Grouser Wheel for Autonomous Enhanced Terrain Adaptability. *Stamps Scholar Summit*, Atlanta, GA, USA, March 2025.

PATENTS

[1] **V. Griffo**, Adaptive Grouser Wheel for Planetary Rovers. *US Provisional Patent Application* No. 63/903,602, filed 22 October 2025. GTRC Ref. 2026-079 (GTRC2026079PRV). United States.

PROFESSIONAL EXPERIENCE

Boston Dynamics, Waltham, MA *July 2026 – Present*
ATLAS Systems Test / Mechanical Engineering Co-Op

- Designing and developing custom test rigs to evaluate the dexterous manipulation capabilities of Atlas' end effectors, validating tasks including lifting, reaching, and fine tool manipulation.
- Performing structural analysis and finite element analysis (FEA) to ensure test fixtures exceed required load and stiffness margins, preventing the test apparatus from becoming the failure mode during evaluation.
- Integrating load cells, force/torque sensors, data acquisition systems, and signal conditioning electronics to quantify gripper performance across representative manipulation tasks.
- Designing and validating leg test fixtures to characterize Atlas' lower-limb performance throughout multiple stages of the walking lifecycle using principles of legged robot dynamics.
- Developing mechanical designs, engineering drawings, and production-ready documentation using CATIA V6 while collaborating with multidisciplinary hardware, controls, and test engineering teams.

MIT Lincoln Laboratory, Lexington, MA *May 2024 – August 2024*
Interceptor Technology and Systems Architecture Researcher

- Designed and developed a low-cost counter-UAV interceptor, reducing manufacturing time and costs via 3D printing and fastener-free design, set for implementation in operational missions for rapid threat response
- Designed, prototyped, and flight-tested a custom 3D-printed high-torque steering control assembly for a low-cost counter-unmanned aerial system (C-UAS) interceptor missile, enhancing maneuverability and control precision
- Designed and manufactured a deployable fin system utilizing quaternions and eigenvalue decomposition, achieving a 60% reduction in the stowed rocket volume, allowing compact stacking and launching of the interceptors
- Helped develop key subsystems, including PCB avionics, and integrated YOLO-based computer vision for real-time target detection, enhancing autonomous threat response for operational deployment
- Implemented physics-based models using MATLAB and MODTRAN to conduct trade studies of electro-optical/infrared (EO/IR) sensors for unmanned vehicles under adverse atmospheric conditions

Georgia Tech Undergraduate Teaching Assistant, Atlanta, GA *Fall 2023, Fall 2024*
Department of Mathematics Linear Algebra Teaching Assistant

- Independently instructed over 40 students in linear algebra, developing comprehensive lecture materials focused on deepening student understanding of key concepts such as eigenvalues, matrix transformations, and vector spaces
- Collaborated with a team of 20+ TAs and professors to coordinate structured grading rubrics and ensure consistent instruction through clear communication and alignment on course objectives and evaluation criteria

COMPETITION TEAMS

URC Mars Rover Team *Member: September 2022 – Present*
Former Chief Engineer / Present Advisor

- Oversaw the design of an autonomous rover consisting of a rocker-bogie drivetrain, 5-DOF robotic arm, precision end-effector, and carbon fiber chassis for simulated Martian navigation in the University Rover Challenge
- Designed a highly dexterous serial 5-DOF robotic arm using Simscape Multibody analysis, validating rigid-body dynamics and torque requirements for manipulation of a 5 kg payload at 1 m reach
- Designed and fabricated a compliant end-effector to manipulate objects with uncertain geometries, improving operational dexterity across competition tasks
- Led the design and manufacture of a custom carbon fiber drone frame and control scheme, featuring a PID-controlled camera gimbal system integrated with a CUAV X7 flight controller for stabilization and flight control
- Designed and manufactured a 2-DOF robotic arm for the URC drone, enabling manipulation and transport of hand tools for mission-critical operations

Georgia Tech Robotics Competition *August 2023 – November 2023*

- Competed against over 200 mechanical engineering students in the Georgia Tech robotics competition, designing and programming an autonomous robot within 5 weeks
- Led the development of a telescoping lift, custom gear drivetrain, high-traction SLA tires, and a passive geometry ball filter, ultimately achieving first seed before the final competition out of 67 teams

VOLUNTEER WORK

Invention Studio Makerspace | *Electronics Apprentice* *September 2022 – March 2024*

- Instructed and guided students and makers in the use of various machining tools and techniques, supporting both research and personal projects within the makerspace
- Demonstrated proficiency in circuit design, electromagnetism, microcontrollers, and shop management by creating complex electronic projects, such as a Gauss Accelerator, and managing the budget to order and restock makerspace components

Skills

- **Technical Software:** SOLIDWORKS, SOLIDWORKS FEA, Siemens NX, Fusion 360, ANSYS, MATLAB, Simulink, Simscape Multibody, MuJoCo, Mission Planner, PX4, Jetson SDK, Linux
- **Hardware & Prototyping:** Precision machining (lathe, mill, waterjet, PCB mill), additive manufacturing (FDM/SLA), TIG/MIG welding, soldering and wiring, sensor integration, rapid prototyping, electromechanical assembly
- **Programming & Embedded Systems:** C++, Python, MATLAB, C#, Java, JavaScript, ROS, Arduino IDE, Raspberry Pi OS, LaTeX
- **Robotics & Control:** Robot kinematics and inverse dynamics, Jacobian-based analysis, legged locomotion, state estimation, actuator design and sizing, PID and field-oriented control (FOC), embedded serial interfaces, hardware-in-the-loop (HIL) control, data acquisition